Mohammed Nagi

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RESEARCH PROFILE

Google DeepMind Scholar in the AIMS MSc in Artificial Intelligence for Science, with a background in electrical and electronics engineering. I use sparse autoencoders to study the internal representations of transformers trained on instrument data rather than text, and I build Human-AI collaboration systems that learn when a model should defer to a human expert. I want to extend both lines of work in a setting where interpretability claims can be tested against the domain knowledge of the scientists who produced the data.

EDUCATION

MSc Mathematical Sciences, specialisation in Artificial Intelligence for Science Sep 2025 – July 2026
African Institute for Mathematical Sciences (AIMS) South Africa *Cape Town*

- **Overall mark:** 84 (Distinction).
- **Relevant coursework:** Applied Machine Learning at Scale, Bayesian Inference, Monte Carlo Methods for Machine Learning, Explainable AI (XAI), Reinforcement Learning, Responsible AI, CUDA Programming for NVIDIA GPUs, Natural Language Processing and Large Language Models.

BSc (Honours) Electrical and Electronics Engineering 2017 – Aug 2025
University of Khartoum *Sudan*

- **Grade:** First Class Honours.
- **Relevant coursework:** Calculus, Linear Algebra, Systems Modelling and Simulation, Signal Processing, Real-Time Embedded Systems.

RESEARCH & TECHNICAL PROJECTS

MSc Thesis: Sparse Autoencoders for Interpretability of InstaNovo | *Python, PyTorch* July 2026

- First sparse autoencoder analysis of a de novo peptide sequencing model: trained SAEs at four encoder depths of InstaNovo over 639,286 spectra, recovering more than 99.5% of sequencing loss at every layer, with chemical properties emerging at the depths fragmentation theory predicts.
- Labelled every peak with 50 concepts computed from fragment-ion theory, then showed the limits of that vocabulary: theory labels only 38% of peaks, and features responding to the rest are described *better* than the strongest concept-associated ones (median r 0.82 against 0.65), recovering real chemistry the annotation omits.
- Two masses recovered from unlabelled peaks match modified-residue immonium ions to within a millidalton, chemistry the annotation could not express; the remaining recurring masses are open candidates for mass-spectrometry validation.

Fairness-Constrained Bandit Routing for Human-AI Triage | *PyTorch, SHAP, Fairlearn* 2026

- Architected PsychRoute, a fairness-constrained human-AI collaboration framework for psychiatric triage (Bipolar Disorder vs Major Depressive Disorder), wrapping a fine-tuned MentalBERT classifier in a LinUCB contextual bandit that routes each encounter across four governance-anchored actions mapped to the Fabric AI governance taxonomy (*Jorgensen et al., 2025*).
- Reduced diagnostic error rate by 35% (0.017 vs 0.026 AI-only and 0.070 human-only) on 3,626 held-out encounters, with formal verification of human-AI complementarity in both directions (*Mozannar & Sontag, 2020*), i.e. the combined system outperforms either party alone rather than merely deferring.

AI-Driven Discovery in Spectral Graph Theory | *Python, PyTorch, RL, GNNs* 2026

- Extended the Cross-Entropy Method (*Wagner, 2021*) with graph neural network policy approximators to discover counterexamples to open conjectures in spectral graph theory, automating search through discrete combinatorial spaces (graphs up to $N = 35$) to surface structural violations of established mathematical bounds.
- Demonstrates machine learning used as an instrument of scientific discovery rather than prediction: the output is a mathematical object a researcher can verify by hand.

Explainable AI & Mechanistic Interpretability | ***PyTorch, SHAP, LIME, TCAV*** 2026

– Implemented axiomatic Shapley decomposition, LIME, and counterfactual explanations from scratch, then contrasted their correlational attributions against Integrated Gradients and Testing with Concept Activation Vectors on Vision Transformers and sequence models, mapping where attribution-based and causal/mechanistic methods agree and diverge.

High-Performance Recommender System (32M MovieLens) | ***Python, Numba, NumPy*** 2026

– Built a hardware-aware Alternating Least Squares pipeline achieving an 8,767× speedup over the Python baseline via JIT compilation and a synchronised CSR/CSC sparse architecture, cutting per-epoch latency from 3.5 hours to 1.46 seconds.

– Verified through latent-manifold analysis that the model autonomously recovers interpretable semantic structure, a representation-integrity audit transferable to other high-dimensional embedding systems.

Agentic Technical Report Generator with Human-in-the-Loop Oversight | ***Python, LangGraph*** 2025

– Built a multi-agent system (Planner, Writer, Validator) with self-correcting validation loops and Pydantic schema constraints to reduce hallucination, exposed through a Gradio interface that makes agent reasoning visible for human monitoring and intervention.

TECHNICAL SKILLS

Languages: Python, CUDA, SQL, MATLAB, LaTeX
ML & Interpretability: PyTorch, Scikit-learn, TransformerLens, Sparse Autoencoders, SHAP, LIME, Integrated Gradients.
Causal & Statistical Methods: Causal Interventions, Feature Ablation, Bayesian Inference, Monte Carlo Methods
Safety, Alignment & Fairness: MC-Dropout Uncertainty, Temperature Calibration, Fairlearn Auditing, Contextual Bandits.
Tools & Platforms: Git, Docker, Google Cloud Platform (GCP)
Research Interests: Mechanistic Interpretability, Machine Learning for the Sciences, Human-AI Collaboration, AI Safety
Natural Languages: Arabic (Native), English (Fluent), French (Beginner)